Eight-Factor Analysis of Natural Kratom Leaf Shows Blanket Bans Could Worsen Public Health

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Author Biography

I am a pharmacist and clinical pharmacology and outcomes researcher.[1] I have written and published extensively about both kratom and substances of abuse[2] in the biomedical literature for many years. I believe I have a realistic view of natural kratom leaf's potential harm and potential benefits.

I am not writing this document as a representative of the University of Connecticut School of Pharmacy, where I am employed, but as the Chair of the Kratom Consumer Advisory Council (KCAC).[3] KCAC is an independent board comprised of a clinician-scientist (myself) and members who were hurt or helped by kratom-derived products. The Global Kratom Coalition (GKC) established KCAC, but its members do not vote on our position statements and are not bound to follow or support our statements or informational content.

Issues and Weaknesses Associated With Blanket Bans

The rationale proposed by advocates of blanket bans for all kratom products suffers from shortcomings that prevent an accurate determination of the risk profile of natural kratom leaf products.

1. A conglomeration of one-sided information focused only on potential risks fails to fully and objectively evaluate all relevant data — including data supporting potential health and societal benefits resulting from properly-regulated use of natural kratom leaf products.

2. Efforts to schedule natural kratom leaf are at odds with the determinations of the Department of Health and Human Services (HHS), the Food and Drug Administration (FDA), the Drug Enforcement Administration (DEA), and the World Health Organization (WHO) — without a reasoned explanation for the conflicting conclusion.

3. Advocates of blanket bans focus on the presence of μ-opioid receptor stimulation and — to the exclusion of other, more substantial data — conflate that presence with a risk profile similar to heroin or fentanyl. They do not consider the proven and major pharmacologic differences between natural kratom leaf and mitragynine versus prescription or illicit opioids:

    - An in vitro Ki value that shows markedly weaker μ-opioid binding strength for mitragynine and evidence that some alkaloids in natural kratom leaf block some opioid receptors.
    - No in vitro self-administration signal for addiction.
    - No in vitro reinforcing effect signal for addiction.
    - Human pupil constriction that is markedly less than with opioids — even at the highest doses — showing a different intensity of pharmacologic effect.
    - No in vitro concerning respiratory depression.

4. Advocates of blanket bans assume the alternative to natural kratom leaf use is for consumers to abstain from all substances, which the available evidence suggests is untrue:

    - Kratom is being used as a substance of abuse substitute or for intractable pain syndromes in people without traditional care options.

    - As described by HHS, viewing natural kratom leaf use in black and white terms — without consideration of context or nuance — can potentially harm public health. Banned substances cannot be utilized in cases where the benefit would outweigh the risk, and they cannot be researched further to better understand benefits and risks.

5. To suggest the available data equates the addictive and safety profile of natural kratom leaf with illicit opioids, cocaine, and methamphetamines is unfounded and contrary to scientific evidence.

6. In lieu of scheduling/banning natural kratom leaf, a Kratom Consumer Protection Act can be enacted that protects consumers but still allows access to products for which the scientific data does not support a complete ban.

Blanket Bans Break With Leading Health Agencies

Concentrated synthetic 7-OH, concentrated synthetic mitragynine pseudoindoxyl, and synthetic MGM-15 and MGM-16 are new to the market, sold as kratom, and driving a surge of addictions across the United States. These synthetic products have a markedly different risk profile — one that aligns with traditional opioids in terms of potency of μ-opioid binding, addiction potential, and risk of respiratory depression in animal models. Efforts by states and municipalities to ban these select kratom-derived products align with pharmacologic evidence and the recommendations of the FDA.

Unfortunately, the fervor generated from these isolated, concentrated synthetic products has led some states and municipalities to ban all kratom products, including natural kratom leaf. This determination is at strong odds with eight-factor analyses[4] conducted in the past[5] and goes against the recommendations of HHS and the FDA. When the FDA recommended that the DEA schedule concentrated synthetic 7-OH in 2025, agency leaders made it clear they were not recommending the scheduling of natural kratom leaf:[6]

> FDA NEWS RELEASE
>
> **FDA Takes Steps to Restrict 7-OH Opioid Products Threatening American Consumers**
>
> *Agency alerts health care professionals and consumers of 7-hydroxymitragynine risks*
>
> **For Immediate Release:** July 29, 2025
>
> The U.S. Food and Drug Administration today is taking a bold step to protect Americans from dangerous, illegal opioids by recommending a scheduling action to control certain 7-hydroxymitragynine (also known as 7-OH) products under the Controlled Substances Act (CSA).
>
> The FDA is specifically targeting 7-OH, a concentrated byproduct of the kratom plant; it is not focused on natural kratom leaf products. 7-OH is increasingly recognized as having potential for abuse because of its ability to bind to opioid receptors. The FDA is releasing a new report to educate the public about the health concerns of 7-OH and its distinction from the kratom plant leaf.

The FDA's report[7] from 2025 identifies why concentrated synthetic 7-OH is not the same as natural kratom leaf — outlining why the agency believes the data is strong enough to state the risks associated with concentrated synthetic 7-OH warrant scheduling while natural kratom leaf does not. Some advocates of blanket bans selectively quote from the FDA's website to create the impression that HHS and the FDA would support scheduling natural kratom leaf. However, they leave out that the former Assistant Secretary of Health, Dr. Brett Giroir, sent a "Kratom Scheduling Rescission Letter"[8] to the acting DEA Administrator in 2018. In that letter, the Assistant Secretary made clear that:

- "I am rescinding our prior recommendation … that mitragynine … be permanently controlled in Schedule I of the CSA."

- "This decision is based on many factors, in part on new data, and in part on lack of evidence, combined with an unknown and potentially substantial risk to public health if these chemicals were scheduled at this time."
- "I am also concerned about the stifling impact of scheduling kratom on our ability to conduct research."
- The Assistant Secretary also wrote that scheduling natural kratom leaf would lead to:
- "... kratom users switching to highly lethal opioids ... risking thousands of deaths."
- "... inhibition of patients discussing kratom use with their primary care physicians leading to more harm, and enhancement of stigma."

The FDA's recently-completed patient tolerability clinical trial,[9] coupled with the agency's explicit statement regarding scheduling concentrated synthetic 7-OH but not natural kratom leaf, shows that its position and that of HHS has not changed. The benign results of the FDA's tolerability study, even at high doses, are now prompting the agency to conduct an additional study which will further our understanding of this natural ingredient.

The FDA is endeavoring to better understand natural kratom leaf and is not concerned about severe, immediate, negative health risks to the public from natural kratom leaf products. The DEA and the WHO[10] have also intensively evaluated whether natural kratom leaf should be scheduled/banned; both agencies have decided against it in the past and have no plans to reassess the issue in the future.

Eight-Factor Analyses Fall Short as Basis for Blanket Bans

Factor 1: Actual or Relative Potential for Abuse

Some advocates of banning natural kratom leaf rely heavily on the FDA Public Health Assessment via Structural Evaluation (PHASE)[11] protocol to characterize whether different alkaloids in natural kratom leaf would interact with opioid receptors, including the μ-opioid receptor.

According to the FDA authors of this study, they developed the PHASE protocol "*as a tool for characterizing newly identified substances that lack in vitro and in vivo data.*" The FDA has transparently stated that it is not the definitive model, but rather a method to attain an initial assessment when in vitro and in vivo data do not exist. In the time since the PHASE assessment was conducted, the alkaloids in natural kratom leaf have been rigorously assessed — both in vitro and in vivo — for actual or relative potential for abuse.

When in vitro and in vivo data diverge from the PHASE assessment, the effective dispositive value should be diminished, as well. In an analytical framework, some types of data are inherently stronger than others and therefore more likely to reflect what will occur in real-life situations. Modeling is not as strong as actual in vitro data, and in vitro data is not as strong as in vivo animal studies.

Similarly, in vivo animal studies are not as strong as observational controlled studies, which are not as strong as clinical trials. Anecdotal experiences (such as case studies, case series, and descriptive studies) are methodologically weak, as there are many biases that can explain an outcome unrelated to the product under assessment.

In Vitro Data

Ki values, which show how strongly a substance binds to a specific receptor, are a valuable and quantifiable metric to determine receptor-binding strength in vitro. In a recent comprehensive review,[12] research experts who conducted sentinel work on natural kratom leaf specified that:

- *"In vitro pharmacological screening using a panel of 82 central nervous system drug targets has been conducted for all the major kratom alkaloids. Due to its natural abundance, mitragynine was hypothesized to interact primarily with opioid receptors, and it has been shown to bind most strongly to the μ-opioid receptor (Ki=709 nM) as a partial agonist. In contrast, it is a weak competitive antagonist at κ-opioid (Ki=1,700 nM) and δ-opioid (Ki=6,800 nM) receptors. Interestingly, mitragynine does not have a high affinity or preference for any one class of receptors and exhibits pharmacological actions through a variety of targets including opioid, serotonin, and adrenergic receptors."*

- *"Speciociliatine interacts primarily with opioid receptors and has analgesic actions in some animal models but not in others, indicating possible pharmacological differences among species. Speciogynine and paynantheine (which only differ by the location of a single carbon-carbon double bond) interact strongly with serotonin receptors, while also interacting moderately with opioid receptors, and to a lesser extent adrenergic receptors. Paynantheine is among the more abundant alkaloids, presenting with mild conditioned place aversion and blocking morphine antinociception at low doses, which may indicate partial antagonist effects at the μ-opioid and partial agonist effects at the δ-opioid receptors. Corynantheidine (which only differs from mitragynine by lacking an -OCH3 group) binds strongly to alpha-adrenergic receptors and has weaker interactions at opioid (Ki=57 nM at μ-opioid receptor) and serotonin receptors."*

- *"The minor alkaloids mitraciliatine and isopaynantheine induce antinociception in animal models that is primarily mediated through κ-opioid receptor activation and do not appear to cause respiratory depression even at very high doses."*

This review, conducted by kratom researchers in the field and peer-reviewed before publication, relied on actual in vitro data that painted a picture far more granular and nuanced than when the PHASE conformational model was used in isolation. It is important to focus on three major findings from these in vitro studies in a rigorous eight-factor analysis.

1. The higher the Ki value, the weaker the binding to the opioid receptor. The Ki values of prescription or illicit opioids[13] (such as oxycodone, morphine, hydromorphone, fentanyl, and sufentanil) range from 0.2 to 25 nM (potent binding) as compared to the Ki value for mitragynine for the μ-opioid receptor of 709 nM[14] (weak binding). The Ki value for the μ-opioid receptor chosen for the expert review was selected because it is conservative, with a Ki value for the μ-opioid receptor of 7,709 nM found in another in vitro study. It is indisputable that the binding of mitragynine to the μ-opioid receptor is far weaker than that of commonly used prescription or illicit opioids.

2. The aforementioned expert review specifies that several of the alkaloids in natural kratom leaf actually block opioid receptors. This is evidence of a complex pharmacologic profile, where the opioid effects of some alkaloids are attenuated by the actions of others. That does not occur with any full-agonist prescription opioids or illicit opioids.

3. Advocates of blanket bans are silent on the non-opioid effects of natural kratom leaf alkaloids. This is important, because major effects of natural kratom leaf — and of mitragynine — are likely derived from non-opioid pathways.

4. There are visible structural differences between prescription and illicit opioids versus mitragynine, as evidenced in the below excerpt from another eight-factor analysis:[15]

Structurally, MG and 7-OH-MG are unrelated to other natural and synthetic MOR agonists (**Figure 1**). MG and 7-OH-MG are considered indole alkaloids, whereas morphine (a natural MOR agonist) is considered a phenanthrene derivative and fentanyl (synthetic MOR agonist) is a 4-anilidopiperidine. All of these MOR agonists share in common a basic amine group, thus they are all alkaloids. There are over 40 indole alkaloids present in the plant that have been reported to date. 7-OH-MG is present in the leaves of *M. speciosa*, though in quantities that are unlikely to contribute to its pharmacologic effect when taken orally; however, MG is metabolized into 7-OH-MG *in vivo*, and could indeed be considered an active metabolite of oral MG. More research is needed to determine this.

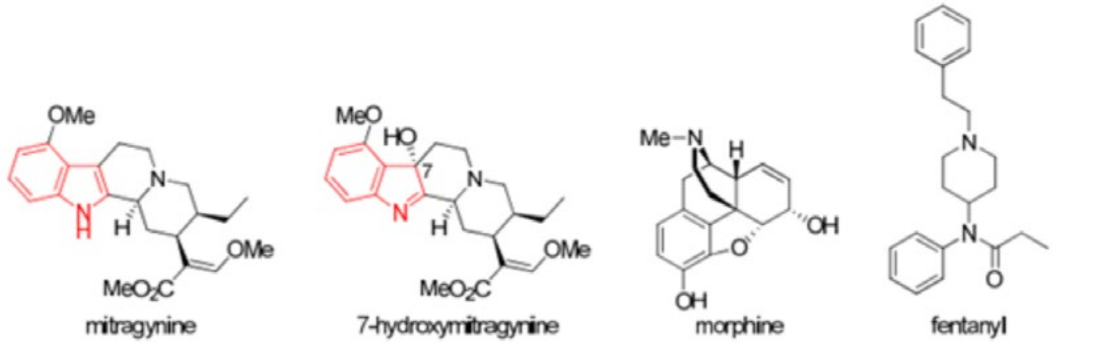


Figure 1. Structures of mitragynine (MG), 7-hydroxymitragynine (7-OH-MG), morphine (a naturally occurring MOR agonist), and fentanyl (a synthetic MOR agonist). The indole group of MG and 7-OH-MG is shown in red.

Mitragynine Does Not Cause Animal Self-Administration

In another eight-factor analysis,[16] there were multiple studies showing the lack of an addiction signal from animal self-administration. Based on intravenous self-administration, there were six studies that found no evidence for reward with mitragynine. Based on intracranial self-stimulation, there were three studies that also found no evidence of reward with mitragynine. These studies establish that mitragynine has distinct pharmacologic differences from prescription or illicit opioids.

Mitragynine Does Not Cause Reinforcing Effects

There are many studies that suggest mitragynine and natural kratom leaf are not the same as prescription or illicit opioids. The lack of a full literature search to identify them is a weakness of blanket ban advocates. While I will not provide an exhaustive list of studies here, I will highlight a few prime examples. However, there is reference to them in updated, peer-reviewed, published eight-factor analyses from Pinney & Associates,[17] the World Health Organization,[18] and academic researchers at the University of Wisconsin.[19]

The following direct quotes are from a critically important mouse study examining the reinforcing effects of mitragynine (referred to as MG) and concentrated synthetic 7-OH (referred to as 7-HMG):[20]

- *"7-HMG, but not MG, substituted for morphine self-administration in a dose-dependent manner in the rat self-administration paradigm. Following the substitution procedure, re-assessment of morphine self-administration revealed a significant increase following 7-HMG and a significant decrease following MG substitution. In a separate cohort, 7-HMG, but not MG, engendered and maintained intravenous self-administration in a dose-dependent manner."*

- *"The present results are the first to demonstrate that 7-HMG is readily self-administered, and the reinforcing effects of 7-HMG are mediated in part by μ and δ opiate receptors. In addition, prior exposure to 7-HMG increased subsequent morphine intake whereas prior exposure to MG decreased morphine intake."*

- *"The present findings indicate that MG does not have abuse potential and reduces morphine intake, desired characteristics of candidate pharmacotherapies for opiate addiction and withdrawal, whereas 7-HMG should be considered a kratom constituent with high abuse potential that may also increase the intake of other opiates."*

This mouse study shows the substitution of morphine with mitragynine does not reinforce opioid addiction, but rather leads to a lower morphine utilization dose over time. These pharmacologic effects are not seen with concentrated synthetic 7-OH, the lab-made opioid that the FDA has sought to have the DEA designate as a Schedule I substance.

Advocates of a blanket ban including natural kratom leaf argue that since a portion of mitragynine can be converted into 7-OH, it must assume the same risk profile. However, this mouse study proves their assumption is unfounded and not pharmacologically accurate. A small portion of the mitragynine introduced into the mice was undoubtedly converted into 7-OH by the body, but even so, it did not change the differential pharmacologic findings.

The results are similar to those obtained in a combination in vitro and in vivo study. In the in vitro portion, mitragynine was found to have μ-opioid receptor specificity that was ~200-fold less than that for morphine. In the in vivo portion, rats were trained to self-administer methamphetamine (0.022 mg/kg/injection) during one-hour daily sessions. They then underwent periodic self-administration of mitragynine, heroin, methamphetamine, or saline.

In rats trained to self-administer methamphetamine, saline substitutions and mitragynine substitutions significantly decreased the number of responses, whereas different doses of methamphetamine (0.002–0.068 mg/kg/injection) or heroin (0.001–0.03 mg/kg/injection) maintained self-administration. Equally as important, pre-session mitragynine treatment (0.1 to 3.0 mg/kg) decreased response rates maintained by heroin, suggesting an attenuation of heroin's negative addictive effects. This is not surprising, since mitragynine is a partial μ-opioid receptor stimulator, not a full opioid receptor stimulator.

WHO Assessment of Natural Kratom Leaf and Mitragynine

The WHO assessment admittedly produced in vitro data showing mitragynine is a μ-opioid receptor and that some of its effects are blocked by traditional opioid receptor blockers. However, there are a number of studies referenced in the WHO report that show major differences from traditional opioids that should be noted. The following are relevant excerpts from the WHO analysis:

- *"... a methanolic extract of kratom leaves (50-300 mg/kg i.p.) does not generate conditioned place preference ... nor does lyophilized kratom tea (100 mg [equivalent to 0.74mg mitragynine]/kg, 1 g [7.4 mg mitragynine]/kg orally) in mice."*

- *"Mitragynine is not self-administered by rats (25-150 μg or 0.1-3.0 mg iv) ... and does not alter the reward threshold (lowering threshold is rewarding) or response latency for intracranial self-stimulation (ICSS) at 1-30 mg/kg ip or intragastric."*

- *"Mitragynine increased the reward threshold at the highest dose studied (56 mg/kg ip), suggesting an aversive action."*

- *"7-Hydroxymitragynine is perceived as more morphine-like than is mitragynine ... Kratom methanolic extract (100 mg/kg p.o., 4.4% mitragynine by weight) did not acutely increase dopamine concentration in mouse brain nucleus accumbens, while a reinforcing dose of alcohol did ... Increasing nucleus accumbens dopamine concentration is an acute effect of almost all substances of abuse, including opioids."*

Mitragynine Does Not Produce Traditional Opioid Respiratory Depression Risks

Another way to determine if the effects of mitragynine are analogous to prescription or illicit opioids is to assess a known, serious, and common adverse event that they cause: respiratory depression. We summarized this assessment in a KCAC position statement.[21]

The most recent study on the subject is special, because it looked at respiratory rate (how many breaths per minute) and minute ventilation (how much air is breathed in and out each minute).[22] This study found that rats given concentrated synthetic 7-OH, like those given morphine, had significantly lower respiratory rates and minute ventilation 30-minutes after dosing.

The study also found that as the dose of concentrated synthetic 7-OH or morphine increased, the extent of respiratory depression got worse. The degree of respiratory depression was very similar when concentrated synthetic 7-OH or morphine was given.

Interestingly, mitragynine administration across all the doses did not suppress respiratory rate or minute ventilation. Both concentrated synthetic 7-OH and morphine decrease respiration through opioid-mediated pathways, as shown by the fact that the effects were reversed by naloxone, an opioid receptor antagonist. On the other hand, the effects of mitragynine on respiration were unaffected by naloxone administration, indicating that mitragynine is not modifying respiration through opioid dependent pathways.

In a mouse study,[23] concentrated synthetic 7-OH (1.9 mg/kg), morphine, (3.8 mg/kg), and mitragynine (5.5 mg/kg) reduced respiration to the same extent in low doses. But as the dose of concentrated synthetic 7-OH and morphine escalated to 10 mg/kg and 30 mg/ kg, the respiratory depression continued to worsen while mitragynine's respiratory depressant effects hit a "ceiling effect" and did not get any worse as the dose rose to 10 mg/kg, 30 mg/kg, or even 90 mg/kg. The extent of respiratory depression with concentrated synthetic 7-OH and morphine across doses were similar.

In a rat study,[24] mitragynine's impact on respiratory depression versus oxycodone was also assessed. Oxycodone administration produced significant dose-related respiratory depressant effects while mitragynine did not.

The best, most current evidence suggests a variety of opioids, including concentrated synthetic 7-OH, induce substantial respiratory depression at higher doses while mitragynine is uniquely lacking in this dose-related phenomenon. Respiratory depression is a major cause of tens of thousands of opioid overdose deaths each year in the United States, so this should be a major consideration.

A proper eight-factor analysis should therefore explore why this stark pharmacologic difference is occurring — and how it suggests that mitragynine in natural kratom leaf is one of many alkaloids with complex overlapping and blocking effects. Clearly, pharmacologic effects cannot be directly extrapolated from other opioid datasets.

Human Data

In considering potential for abuse, it is critical to understand that addiction, now termed a use disorder, is determined by a number of factors — some of which are much more detrimental to individual and population health than others. I believe this is the reason why HHS and the FDA have taken a prudent and nuanced approach to natural kratom leaf versus concentrated synthetic 7-OH products. All states and municipalities should take a similar approach that strives to better and more deeply understand all the components of addiction and its nuances, particularly in the context of a complex substance like natural kratom leaf.

Pupil Dilation and Addiction Potential

There are two gold-standard studies (placebo-controlled clinical trials) that have assessed the tolerability of single-dose kratom use. One of the two also assessed multiple-day therapy. The FDA conducted the first of these clinical trials.[25] In the trial, normal volunteers were given placebos, or one of several doses of natural kratom leaf (1-12 g, with 12 grams being four times higher than the normally consumed dose).

The FDA found that natural kratom leaf at high doses produced relatively mild (~2 mm) pupillary constriction compared to morphine and oxycodone, which produced >4 mm of pupillary constriction at high doses. This shows another clear μ-opioid receptor effect that is less than half as pronounced with natural kratom leaf than with prescription opioids.

The FDA evaluated four markers that could suggest future addictive potential: feeling high/euphoric, feeling drunk, drug liking, and desire to take the substance again. These factors were not elevated above placebo levels for normal consumption doses of 1-3 g. There were some qualitative increases at 8-12 g doses, but the FDA did not identify these as concerning and merely recommended additional study. The FDA stated:

- *"Kratom did not appear to produce clear, dose-related or significant effects on VAS-rated measures commonly associated with abuse potential including ratings of drug liking, overall drug liking, and assessments of [taking the] drug again."*

- *"These data warrant additional studies to more thoroughly assess the abuse potential of kratom."*

If the FDA concluded there is no clear signal of risk in humans and that the correct step to protect public health is to conduct additional studies (one of which the FDA is already planning), it is difficult to reconcile why advocates of a blanket ban are asserting there is currently sufficient evidence to schedule natural kratom leaf due to its addictive potential.

What the FDA study clearly identified was dose-related increases in gastrointestinal adverse events (such as nausea and vomiting). This effect, bolstered by the leaf vegetal matter, is a built-in hedge against binging the product and one that prescription and illicit opioids do not have.

In a separate, multiple-dose clinical trial,[26] investigators gave 116 people placebos or 500 mg, 1 g, 2 g, and 4 g of kratom, with a mitragynine concentration per gram that was higher than that of the FDA tolerability study. The researchers also evaluated euphoria and feeling drunk. They found no more than one person reporting feeling euphoria per dosing group and in the placebo group. They found no people feeling drunk in the placebo or 500 mg to 2 g group, but two in the 4 g group. They did not identify natural kratom leaf's effect on either marker as a concerning signal of addiction potential. These placebo-controlled dose finding clinical trials have the highest internal validity of all study types.

Many advocates of blanket bans disregard clinical trial evidence and cling to the weakest form of evidence: the case report. What makes the case report so weak is that there is no control group, there are limited assessments for confounding factors, there is no ability to determine comparative risks, and there is limited ability to tease out risks from natural kratom leaf versus concentrated synthetic kratom-derived products.

It is highly likely that a case of "kratom" harm that was reported[27] where a free sample[28] resulted in an alleged addiction was a concentrated synthetic 7-OH product and not a natural kratom leaf product, given the stated marketing of manufacturers and distributors of concentrated synthetic 7-OH. As such, this case was likely misattributed to natural kratom leaf.

Natural Kratom Leaf Reducing Use of Substances of Abuse

Research suggests natural kratom leaf may be beneficial to those who are addicted to other substances. In a cross-sectional study[29] of 332 people who used natural kratom leaf for any reason, 59% used it as a substitute for heroin and methamphetamine.

In another cross-sectional assessment, 961 patients[30] used kratom along with another substance of abuse specifically to manage their disorder. Only 5.2% of people still used the other substance within six months of starting kratom (94.8% stopped). Importantly, 83% of people stopped using the other substance of abuse for >1 year (this includes 19% stopping for 1-2 years, 29% for 2-5 years, and 35% for >5 years).

In a much larger cross-sectional study,[31] over 7,600 people were asked about why they used natural kratom leaf. Overall, 2,352 respondents wanted to either stop or wean down use of an opioid they were addicted to, and 4,811 had untreated pain. Importantly, 3,715 of 7,605 (49%) people reported eliminating their opioid use or reducing their reliance on opioids.

Natural Kratom Leaf Reducing Illicit Drug Adverse Events

Similarly, there is research indicating natural kratom leaf use may help reduce adverse events associated with users' abuse of other substances. For instance, there are three studies assessing the impact of kratom substitution on adverse events caused by illicit opioids.

The first study[32] was a cross-sectional evaluation of 163 participants with a history of illicit opioid use that was replaced by natural kratom leaf use. Researchers found that natural kratom leaf substitution was associated with a decreased prevalence of slowed or stopped breathing (respiratory depression), physical pain severity, cravings, constipation, insomnia, depression, loss of appetite, decreased sexual performance, weight loss, and fatigue associated with illicit opioid use.

The next two cross-sectional studies were specifically focused on HIV risk behaviors (the same behaviors that cause hepatitis B and C infections, as well as bacterial and fungal infections). In the first of these studies,[33] 260 participants who attempted to substitute natural kratom leaf for illicit drugs were assessed. Participants reported a reduction in illicit use of heroin, methamphetamine, amphetamine, cannabis, benzodiazepines, ketamine, methadone, and alcohol after natural kratom leaf initiation and a reduction in HIV risk behaviors (such as injecting illicit drugs and sharing needles and syringes). In the second of these studies,[34] 32 HIV-positive users dramatically reduced their illicit opioid injection and reduced sharing syringes and needles after using natural kratom leaf.

Natural Kratom Leaf and Fulfilling Societal Obligations

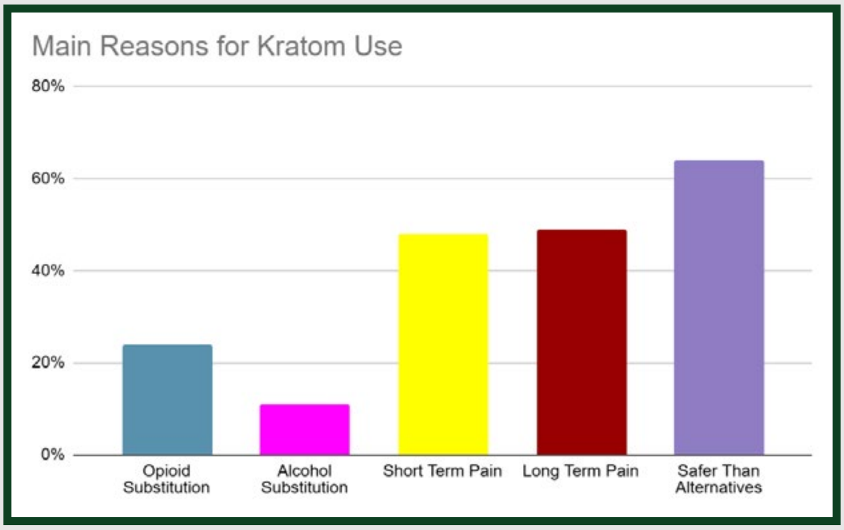
In addition, there is research indicating natural kratom leaf use actually aids users' societal functionality. There are three descriptive datasets assessing the ability of people to fulfill their daily obligations when using natural kratom leaf. In the first descriptive study of 103 users of natural kratom leaf,[35] 54% of users said that natural kratom leaf helped them meet their daily obligations while only 3% felt it impeded them meeting their daily obligations. The rest of the respondents felt the effects were more equivocal or did not express an opinion.

In a new study,[36] 16 people reported a pronounced reduction in the frequency and quantity of alcohol use when they started using natural kratom leaf to manage their issue. While alcohol affected their daily functioning, they reported natural kratom leaf did not do so.

Another recent study of 357 kratom users (almost all natural kratom leaf) provided a strong assessment of this trade-off between potential benefit and risk.[37] Group A had the highest daily dose of kratom while E had the lowest. When asked if kratom was beneficial or harmful, respondents had differing views:

- >88% of participants found kratom was a real benefit
- >88% felt it was therapeutic
- ~63% felt it was lifesaving
- ~17% thought it was addictive
- ~4% felt use was problematic
- ~3% felt use was a burden

Although there was mixed reaction, the vast majority of people felt kratom use benefitted them and a minority found it harmed them. A closer examination of who these people are is instructive. When asked why they were using natural kratom leaf, this was their response:



The natural kratom leaf users had a high likelihood of having used opioids, amphetamines, cocaine, alcohol, or hallucinogens in the past. When asked about the five most impactful substances they chose from the past, this is what they said:

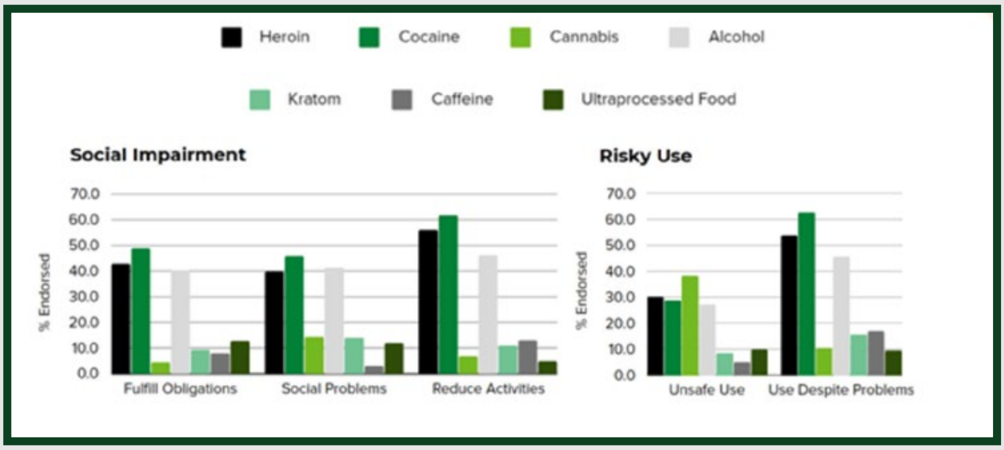
- Alcohol: 25-50%
- Non-prescription marijuana: 25-45%
- Prescription opioids: 25-38%
- Hallucinogens: 13-30%
- Non-prescription opioids: 13-25%
- Heroin: 10-25%
- MDMA: 10-25%
- Prescription marijuana: 10-25%
- Amphetamines: 8-20%
- Cocaine: 5-13%

When asked about how natural kratom leaf impacts their functioning, 70-90% said it improved while 5% found natural kratom leaf a hindrance. While some people's lives were worse off after using natural kratom leaf, a strong majority of people's lives were better after using natural kratom leaf for their intractable pain or as a substitute for other substances of abuse.

A GKC assessment[38] provides useful information about how people using illicit opioids fare versus those using natural kratom leaf or mitragynine extract products. Overall, 86% of people using heroin met the criteria for substance use disorder versus 29% for natural kratom leaf. Of the population with a substance use disorder, 62% versus 29% had a severe disorder (53.3% versus 8.4% of the total) associated with heroin versus natural kratom leaf.

The biggest differences between heroin use disorder and kratom use disorder were seen in the percentage of people unable to fulfill family, work, and societal obligations (42% versus 9%), causing societal problems (40% versus 12%), demonstrating unsafe use practices (30% versus 8%), and using despite problems (family, work, and societal; 53% versus 15%).

People were less likely to have unsuccessful attempts at quitting natural kratom leaf (73% versus 33%), fewer cravings (62% versus 30%), less tolerance (63% versus 39%), and fewer withdrawals (60% versus 33%) than with heroin. The dramatically more negative impact of cocaine and alcohol on people's ability to function also suggests that switching from these other substances of abuse to natural kratom leaf would provide societal benefits, as demonstrated in the below figure:



Taken together, the totality of data related to Factor 1 of the eight-factor analysis shows that natural kratom leaf is not analogous to traditional prescription of illicit opioids and is different than concentrated synthetic kratom-derived products.

Factor 2: Scientific Evidence of Pharmacological Effect

As discussed above, the pharmacologic effects of natural kratom leaf are characteristically different in many ways:

- Partial agonist and much weaker binding to opioid receptors (unlike morphine, heroin, and fentanyl)
- Alkaloids that block some opioid receptors (unlike morphine, heroin, and fentanyl)
- Effects at non-opioid sites in the body, including serotonin, adrenergic, and antiinflammatory mechanisms (unlike morphine, heroin, and fentanyl)
- Less pupil constriction, no reinforcing effects, and no respiratory depression effects as compared to traditional opioids

Data suggest periods of natural kratom leaf use versus periods of illicit opioid use have less risk of blood-borne illness and result in a better ability to work and fulfill societal obligations (such as people caring for themselves and their families). A weaker dataset suggesting the same is true for people substituting alcohol or stimulant addiction with natural kratom leaf.

There is some data specifically assessing kratom's pain-relieving effects. This data is preliminary and not definitive, but it is positive.[39] In a randomized controlled trial, 26 males were enrolled. Pain tolerance increased significantly one hour after kratom ingestion, and it was unchanged after consuming placebo drinks. No discomfort or signs of withdrawal were reported or observed during 10-20 hours of kratom discontinuation.

In a second study, the results showed nearly half of participants (49.1%) met the criteria for chronic pain, with many experiencing significant pain relief and high effectiveness in pain management through kratom use. A majority (69.2%) reported challenges obtaining adequate pain treatment, which influenced their decision to try kratom. Few participants expressed concerns about overuse or major side effects, indicating pain relief was the most common motivation for daily kratom use, regardless of chronic pain status. There was no significant association between daily pain levels and kratom use frequency, nor any difference between individuals with or without chronic pain. Current kratom use was associated with lower pain levels, and stronger subjective effects were linked to greater pain reduction, with the effect being more pronounced in those with chronic pain.[40]

Factor 3: State of Current Scientific Knowledge

Advocates of blanket bans selectively cull information that supports their preconceived narrative without displaying the totality of the evidence and the relative strength and applicability of that evidence. As delineated above, the current state of the literature regarding natural kratom leaf is that it can hurt some people, but there are more people being helped than hurt by allowing its availability

Factor 4: History and Current Pattern of Abuse

Natural kratom leaf has been consumed since at least the 1830s among the indigenous populations of Southeast Asia. While waves of serious opium and opioid addiction have swept the world numerous times, the same phenomenon has not been associated with natural kratom leaf.

In contrast, in Southeast Asia, natural kratom leaf has been commonly used to treat the ravages of opium and opioid addiction. The main issues in the kratom-derived product marketplace seem tied to the introduction of concentrated synthetic 7-OH, concentrated synthetic mitragynine pseudoindoxyl, and synthetic MGM-15 and MGM-16 products. Some products with mitragynine extracts have been found to be enriched with concentrated synthetic 7-OH. These products have a markedly different risk profile than natural kratom leaf products in the United States, and their reasons for use and patterns of use seem different.

In addition, some natural kratom leaf products have heavy metal contamination or are misleadingly marketed or mislabeled. These are real issues with some producers and sellers, but it does not change the inherently low risk associated with use of unadulterated products made according to good manufacturing practices.

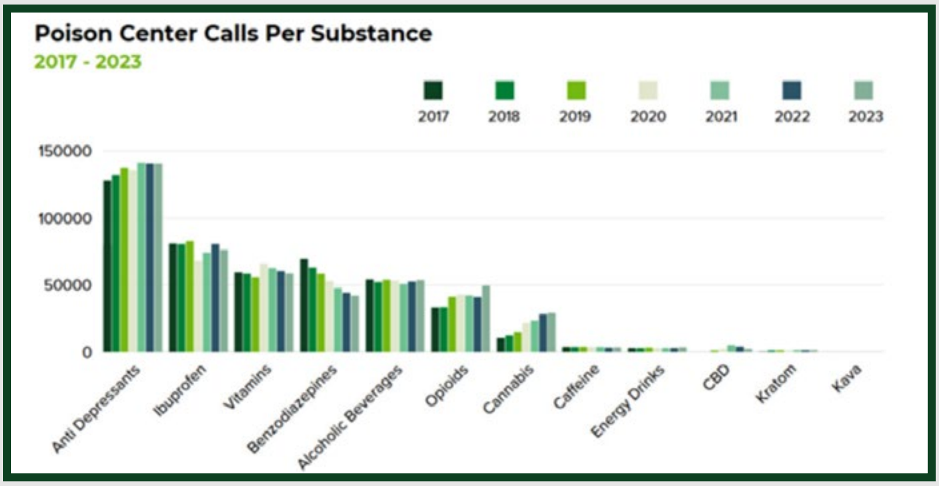
I have personally written numerous articles[41] on adulteration of dietary supplements.[42] Such adulteration is unquestionably a major issue with natural products in the United States. However, if all dietary supplement products were scheduled because a few bad actors were not following the Dietary Supplement Health and Education Act, we would lose the benefit of an exceptional number of natural products.

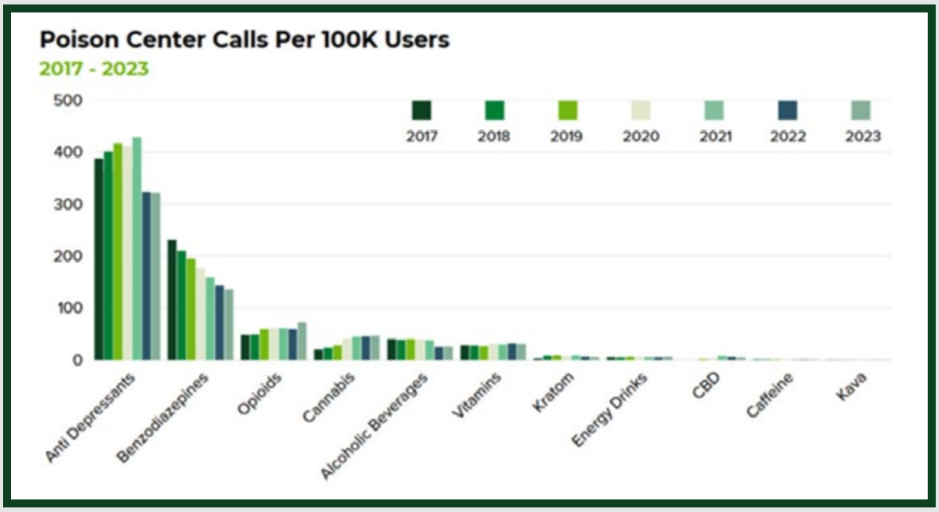
All these quality control issues can be dealt with through a Kratom Consumer Protection Act that is meaningfully enforced. KCAC has many best practices that can support states and municipalities in reducing the risk of unwanted adulteration without banning a substance that many people believe has restored their ability to function and reduced risks associated with illicit opioid use (such as respiratory depression and blood-borne diseases).

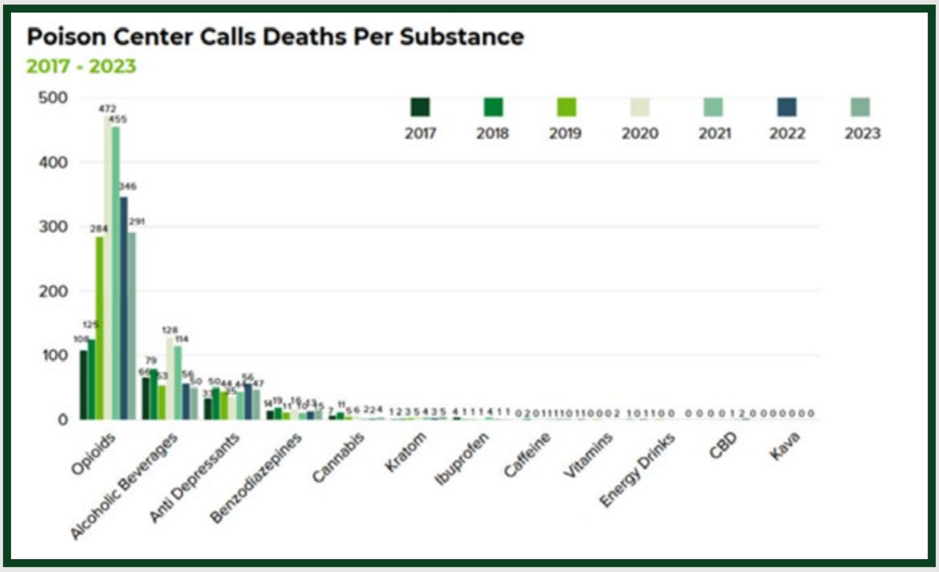
The poison control center data touted by some advocates of blanket bans represents the number of calls associated with all kratom-related products. Until February 2025, the poison control dataset did not include a separate category for concentrated synthetic 7-OH products. Thus, the reported data is tainted with concentrated synthetic 7-OH cases.

Additionally and more importantly, there is no denominator for how many users there were, and no other products are included for the sake of comparison. However, there is an available assessment[43] of the calls per 100,000 users of kratom versus other commonly used substances, including substances of abuse.

Below, I provide several figures from that analysis. The first reflects the number of calls with no denominators. The second reflects the number of calls per 100,000 users. The third reflects the number of deaths per 100,000 users. This data makes clear that the number of calls and deaths are less for natural kratom leaf than for the dangerous substances of abuse which people routinely substitute for kratom. Providing only case numbers that cannot be placed into proper context — rather than conducting a comparative risk assessment — is problematic for public-health decision-making.







Factor 5: Scope, Duration, and Significance of Abuse

Available data show there is a risk of abuse with natural kratom leaf products and that addiction can lead to patient harm. However, the evidence suggests the overall benefits from people moving from illicit substances of abuse to natural kratom leaf seem to be much more impactful to society than the risks of people being harmed. It is therefore important to reduce the risk of harm through regulations — and not through a blanket ban that would potentially hurt a large number of people.

Factor 6: Risk to Public Health

Advocates of blanket bans refer to anecdotal evidence of harm without also providing anecdotal evidence to the contrary and comparatively assessing accounts. What is the true risk to the public compared with the benefits — and how much of that risk can be avoided by requiring independent, third-party testing for heavy metals and good manufacturing practices?

I have provided data that helps answer these questions. Heavy metal contamination is a real concern and one that I have personally highlighted in the past.[44] However, this is not an insurmountable issue that warrants the banning of an entire substance. Rather, it is an issue that could be fixed with third-party analysis. If food or supplement categories were banned because of the potential risk of heavy metal contamination, we would lose all rice products, apples, baby food,[45] cinnamon, protein supplements,[46] and many other products.

Advocates of blanket bans provide data suggesting the number of children who used “kratom” in the past year in the United States was ~45,000. While anything above zero is bad for society, that number is very small when compared to the 134 million who used alcohol, 48 million who smoked, 28 million who vaped, and 44 million used cannabis.[47] Additionally, this number of children using “kratom” does not differentiate between natural kratom leaf and mitragynine extracts or concentrated synthetic products.

Natural kratom leaf can cause gastrointestinal issues given the fibrous vegetal matter that is otherwise removed when making extracts. Those adverse gastrointestinal effects — and users’ desire to avoid them — provide an additional barrier to overusing natural kratom leaf that does not exist for other substances.

Comparative Risk

I believe advocates of blanket bans fail to fully or adequately evaluate all the scientific data regarding natural kratom leaf use or properly engage in a comparative risk-benefit assessment of natural kratom leaf use in the broader context of widespread opioid abuse. In this regard, a few general observations merit additional consideration.

Prescription and Illicit Opioid Use Disorder

According to Avalere Health,[48] 6 million Americans have an opioid use disorder, costing the United States almost $3.98 trillion dollars in 2024. This was driven by lost productivity (on the individual, corporation, and family levels), health care costs, criminal justice costs, societal costs (such as social services for individuals and families), lost money or stolen property, substance treatment costs, and lost tax revenue.

For comparison, the Centers for Disease Control and Prevention (CDC) estimates the economic toll of heart attacks and strokes totals $254 billion in health care costs and $168 billion in lost productivity per year. This is because heart disease usually manifests later in life and does not strain public health services or the criminal justice system to the same extent as opioid abuse.

When lifetime direct costs and specific factors (such as reduced quality of life and lifespan) are factored in, the total lifetime cost to the United States per person with uncontrolled opioid use disorder is $532,000. The use of behavioral therapy with FDA-approved therapies (such as methadone or buprenorphine) can save $271,000 to $295,000 per person over a lifetime by allowing people with opioid use disorder to live longer and more productive lives.

Unfortunately, in the United States, only 22.5% of people are receiving any medication to treat their opioid use disorder.[49] People with a family income below $50,000, males, non-Hispanic whites, and people living in metropolitan areas were more likely to access care. In another study of unhoused people in California,[50] only 6.7% of people were receiving care for substance use disorders. However, 21.2% desired such care but could not access it.

All this data supports the conclusion that vulnerable people with opioid use disorder — without sufficient health care coverage or housing — are at high risk of continuing illicit opioid use and incurring the immense personal and societal costs associated with it.

Below, I identify several categories where opioid use disorder negatively impacts public health. Although natural kratom leaf should not be used by people who can access and afford FDA-approved options (such as methadone, buprenorphine, or naltrexone), the data indicates that natural kratom leaf can serve a positive role in addressing opioid use disorder.

Adverse Health Outcomes From Opioid Use Disorder

Opioid Overdose Deaths

According to the CDC,[51] the United States has made tremendous progress in reducing opioid-related deaths. Overdose deaths involving opioids decreased from an estimated 83,140 in 2023 to 54,743 in 2024. This drop is largely attributed to a public-private partnership focused on dramatically increasing the accessibility of naloxone in communities and training first responders and community members themselves. Increased accessibility to fentanyl test strips is also thought to have reduced inadvertent exposure to fentanyl in illicit or counterfeit products, reducing deaths.

While tremendous progress has been made in addressing deaths, another important component of reducing the public-health risk from illicit opioids is moving people from using those drugs to viable alternatives. There is strong data to support FDA-approved options (such as methadone, buprenorphine, and naltrexone) as illicit opioid substitutes. There is also descriptive data from surveys and anecdotal experiences to support the use of natural kratom leaf products for this purpose.

One important distinction between natural kratom leaf's most abundant alkaloid (mitragynine) versus potent traditional semi-synthetic or synthetic opioids is that mitragynine does not induce dose-dependent reductions in breathing in preclinical studies.[52] This finding is bolstered by years of anecdotal and clinical trial data for both single-dose[53] and multiple-dose (15 days)[54] use at doses up to 12 g and 4 g, respectively.

Hospitalizations and Severe Brain Damage

The CDC estimates that from 2010 to 2020, for every fatal opioid overdose, there were 15 non-fatal ones.[55] That puts the number of opioid overdoses at over 1 million per year. While recent progress has been made in reducing the number of deaths due to enhanced naloxone availability, that availability likely does not reduce the number of overdoses that occurred.

In 2024, the CDC reported non-fatal opioid overdoses caused 43.2 emergency department visits and 17.7 hospital admissions per 100,000 persons.[56] Assuming 330 million people in the United States, this is approximately 142,560 emergency visits and 58,410 hospital admissions due to opioid overdose in the United States annually.

In a separate evaluation of patients hospitalized at 162 hospitals in 44 states for opioid overdose from 2009 to 2015, approximately 17% of patients admitted required intensive care (or about 9,930 intensive care unit admissions based on a total U.S. population of approximately 330 million people).[57] Of the patients with intensive care unit admission:

- 8% experienced anoxic brain injury (or about 794 anoxic brain injuries annually).
    - Developing anoxic brain injury can have immense societal and family implications for those that must provide ongoing care for patients who cannot feed or care for themselves.[58]
- 25% experienced aspiration pneumonia (or about 2,483 aspiration pneumonias annually).
- 15% developed severe muscle breakdown called rhabdomyolysis (or about 1,490 rhabdomyolysis cases annually).

Viral Infections

Injecting illicit drugs annually causes the following new viral infections:[59]

- 2,300 new HIV infections
- 35,048 new Hepatitis C infections (Approximately 70% develop chronic disease and 15% of those cases result in liver failure, liver cancer, or liver cirrhosis.)
- 3,312 new Hepatitis B infections (Approximately 3% develop chronic disease which can cause aforementioned liver issues.)
- 297 new Hepatitis A infections (Approximately 1% develop severe issues.)

These infections are not limited to newly infected people but can spread to others not abusing injected drugs through sexual contact or pregnancy.

Bacterial and Fungal Infections

Over a 10-year period, there was almost a doubling of admissions for serious infections (from 3,421 in 2002 to 6,535 in 2012).[60] This included a:

- 1.5-fold increase in heart valve infections (infective endocarditis)
- 2.2-fold increase in bone infections (osteomyelitis)
- 2.6-fold increase in spinal epidural abscesses

From 1993 to 2010, the national incidence of skin and soft tissue infections due to injected illicit opioids was assessed (<4 per 100,000 in 1993 to >9 per 100,000 in 2010). This was before the use of xylazine in illicit opioid products, which by itself compromises the ability of the immune system to fight skin infections.

Homelessness

In an assessment of homeless or unhoused people in California:[61]

- 10.4% of people were regular opioid users, and 3.7% were occasional opioid users.
- If those statistics are consistent with 650,000 unhoused people nationwide, there would be 91,650 unhoused opioid users in the United States.

Taken together, assessing the risks of illicit opioid use for control of intractable pain or due to an opioid use disorder is critical to understanding the trade-off when people are using natural kratom leaf. Simply banning natural kratom leaf products because there are some people who have been harmed by their use could result in a much larger number of people moving back to illicit opioids and taking on the negative personal, family, and societal harms that ensue.

Why People Do Not Access FDA-Approved Options

There are many reasons people with opioid use disorder do not seek FDA-approved treatment options:

- They may not have insurance coverage for care or have high-deductible plans they cannot afford.
- They may not have a health care provider.
- They may not have reliable transportation or the ability to make scheduled appointments.
- They may not want to disclose to loved ones or employers they have an opioid use disorder.
- They may not have benefitted from available, FDA-approved options.
- They may have experienced intolerable adverse effects from available, FDA-approved options.

As such, there is a great need for vulnerable people to use an accessible and affordable option if the alternative is continued use of illicit opioid products. In a public-health scenario where FDA-approved options are inaccessible or have failed many people, the decision to use natural kratom leaf instead of continuing consumption of illicit opioids could be positive.

Factor 7: Potential of Substance to Produce Psychic or Physiological Dependence Liability

There is some potential to induce psychic and physiologic dependence with prolonged use of natural kratom leaf products. However, given the nature of the population using natural kratom leaf products for pain relief in lieu of opioids or to substitute for substances of abuse that are already negatively impacting people's ability to function, the benefits and harms must be compared with those associated with the substances of abuse that the same people are using or likely to use otherwise.

Factor 8: Whether Substance Is Immediate Precursor

The multi-cannabinoid and terpene-containing natural hemp leaf is not a precursor to tetrahydrocannabinol (THC), even though the cannabidiol (CBD) in cannabis can be chemically manipulated in a laboratory to form THC. Similarly, natural kratom leaf is not just mitragynine — it is a multi-alkaloid substance with a variety of biological actions at multiple receptor types.

While some mitragynine is converted into 7-OH in the body when natural kratom leaf is ingested, all the pharmacologic and clinical evaluations of mitragynine or natural kratom leaf have accounted for that conversion. This fact does not change the risk profile for natural kratom leaf. Specifically, the mitragynine to 7-OH conversion does not change the comparative risk profile of natural kratom leaf versus concentrated synthetic 7-OH products. Research clearly shows the antinociceptive effects of mitragynine are not driven by the conversion to 7-OH62 and suggests a much more complicated pharmacologic picture.[63]

Responsible Regulation Over Blanket Bans

KCAC knows natural kratom leaf use is not risk-free and public-health protections are needed. We have numerous position statements[64] outlining how to regulate natural kratom leaf products to reduce public-health risks without denying vulnerable people access to the products.

One of the most meaningful proposals is pre-registration of products, which would require a photo of their packaging and dosage form. If products are judged to be appealing to children, they can be denied. If products are not properly labeled, they can be denied. If products are overly concentrated with non-intuitive serving sizes, they can be denied. If products are in a form that can not be swallowed (such as vape, sublingual or buccal, or transdermal products),[65] they can be denied.

The pre-registration process would also require that independent, third-party laboratory testing show a lack of adulteration and contamination. If a certificate of analysis is not uploaded for the product, it can be denied. For example, if lab testing shows an unnatural level of 7-OH, mitragynine pseudoindoxyl, or any synthetic alkaloid, the product can be denied. These measures, coupled with statewide purchasing bans for consumers under 21 years of age, can reduce societal risk.

References

1. C. Michael White – The Conversation
2. UConn Pharmacy Research on Kratom and Substances of Abuse
3. About KCAC — Global Kratom Coalition
4. Kratom Abuse Potential 2021: An Updated Eight Factor Analysis - PMC
5. Kratom Document - Eight Factor Analysis - CWC
6. FDA Takes Steps to Restrict 7-OH Opioid Products Threatening American Consumers
7. 7-Hydroxymitragynin (7-OH): An Assessment of the Scientific Data and Toxicological Concerns Around an Emerging Opioid Threat
8. Assistant Secretary for Health Giroir Letter to Acting DEA Administrator Dhillon
9. A Pilot, Placebo Controlled, Dose Finding Pharmacodynamic and Pharmacokinetic Study of Orally Administered Botanical Kratom in Non-Dependent, Recreational Polydrug Users with Opioid Experience Under Fed Conditions - FDA
10. ECDD: Kratom (Mitragyna speciosa), Mitragynine, and 7-Hydroxymitragynine - WHO
11. Evaluating Kratom Alkaloids Using PHASE
12. An Update on the Clinical Pharmacology of Kratom: Uses, Abuse Potential, and Future Considerations - PubMed
13. Uniform Assessment and Ranking of Opioid Mu Receptor Binding Constants for Selected Opioid Drugs
14. Pharmacological Comparison of Mitragynine and 7-Hydroxymitragynine: In Vitro Affinity and Efficacy for μ-Opioid Receptor and Opioid-Like Behavioral Effects in Rats
15. Kratom Document - Eight Factor Analysis - CWC
16. Kratom Abuse Potential 2021: An Updated Eight Factor Analysis - PMC
17. Kratom Abuse Potential 2021: An Updated Eight Factor Analysis - PMC
18. ECDD: Kratom (Mitragyna speciosa), Mitragynine, and 7-Hydroxymitragynine - WHO
19. Kratom Document - Eight Factor Analysis - CWC
20. Abuse Liability and Therapeutic Potential of the Mitragyna speciosa (Kratom) Alkaloids Mitragynine and 7-Hydroxymitragynine - PubMed
21. Have "7" Products With 7-Hydroxymitragynine (7-OH) Been Proven to Be Safe?
22. Mitragynine and 7-Hydroxymitragynine: Bidirectional Effects on Breathing in Rats - PubMed
23. The Respiratory Depressant Effects of Mitragynine Are Limited by Its Conversion to 7-OH Mitragynine - PubMed
24. Respiratory Effects of Oral Mitragynine and Oxycodone in a Rodent Model - PubMed
25. A Pilot, Placebo Controlled, Dose Finding Pharmacodynamic and Pharmacokinetic Study of Orally Administered

Botanical Kratom in Non-Dependent, Recreational Polydrug Users with Opioid Experience Under Fed Conditions - FDA
26. Safety and Tolerability of Single and Multiple Daily Oral Doses of Dried Kratom Leaf Powder in a Randomized Trial in Healthy Volunteers - PubMed
27. Inside the Kratom Industry's 7-OH Tug of War Between Profits and Public Safety
28. Free Sample Tablet - 7-OH
29. The Use of Kratom (Mitragyna speciosa Korth.) Among People Who Co-Use Heroin and Methamphetamine in Malaysia
30. Exploring the Self-Reported Motivations of Kratom (Mitragyna speciosa Korth.) Use: A Cross-Sectional Investigation
31. Patterns of Kratom Use and Health Impact in the US—Results From an Online Survey - ScienceDirect
32. Self-Reported Prevalence and Severity of Opioid and Kratom (Mitragyna speciosa Korth.) Side Effects
33. Self-Report Data on Regular Consumption of Illicit Drugs and HIV Risk Behaviors After Kratom (Mitragyna speciosa Korth.) Initiation Among Illicit Drug Users in Malaysia - PubMed
34. Effects of Kratom (Mitragyna speciosa Korth.) in Reducing Risk-Behaviors Among a Small Sample of HIV Positive Opiate Users in Malaysia - PubMed
35. Searching for a Signal: Self-Reported Kratom Dose-Effect Relationships Among a Sample of US Adults With Regular Kratom Use Histories
36. Kratom (Mitragyna speciosa) as a Replacement for Alcohol Among a Sample of Malaysian Adults with a History of Alcohol Use Problems
37. Ecological Momentary Assessment of Self-Reported Kratom Use, Effects, and Motivations Among US Adults - PMC
38. Understanding Kratom - GKC
39. Kratom and Pain Tolerance: A Randomized, Placebo-Controlled, Double-Blind Study - PMC
40. Kratom (Mitragyna speciosa) Use for Self-Management of Pain: Insights From Cross-Sectional and Ecological Momentary Assessment Data
41. Dietary Supplements Pose Real Dangers to Patients
42. Continued Risk of Dietary Supplements Adulterated With Approved and Unapproved Drugs: Assessment of the US Food and Drug Administration's Tainted Supplements Database 2007 Through 2021 - PubMed
43. Comparative Safety of Kratom Compared to Other Commonly Used Substances - GKC
44. Elemental Impurities (Heavy Metals) in Kratom Products: An Assessment of Published Individual Product Analyses - PubMed
45. Parents Can Soon Use QR Codes to Reveal Heavy Metal Content in Baby Food - UConn Today
46. New Report Finds High Levels of Lead in Some Protein Powders. A Pharmacologist Explains the Data - PBS News
47. Key Substance Use and Mental Health Indicators in the United States: Results from the 2024 National Survey on Drug Use and Health
48. The Cost of Addiction: Opioid Use Disorder in the United States - Avalere Health
49. Use of Medication for Opioid Use Disorder Among Adults With Past-Year Opioid Use Disorder in the US, 2021 - Clinical Pharmacy and Pharmacology, JAMA Network Open
50. Illicit Substance Use and Treatment Access Among Adults Experiencing Homelessness
51. U.S. Overdose Deaths Decrease Almost 27% in 2024 - NCHS Pressroom, CDC
52. Have "7" Products with 7-Hydroxymitragynine (7-OH) Been Proven to Be Safe?
53. Pilot, Dose-Finding Study of Kratom Alkaloids: Study Design Updates
54. Safety and Tolerability of Single and Multiple Daily Oral Doses of Dried Kratom Leaf Powder in a Randomized Trial in Healthy Volunteers
55. Estimating the Ratio of Fatal to Non-Fatal Overdoses Involving All Drugs, All Opioids, Synthetic Opioids, Heroin or Stimulants - PMC
56. DOSE-DIS Dashboard: Nonfatal Overdose Emergency Department and Inpatient Hospitalization Discharge Data
57. The Critical Care Crisis of Opioid Overdoses in the United States - Annals of the American Thoracic Society
58. The Hidden Victims of the Opioid Crisis: The Ones Who Lived - The Guardian
59. Viral Infections Related to Injection Drug Use - PMC
60. Infectious Complications in Injection Drug Use - PMC
61. Illicit Substance Use and Treatment Access Among Adults Experiencing Homelessness
62. 7-Hydroxymitragynine Is an Active Metabolite of Mitragynine and a Key Mediator of Its Analgesic Effects - PMC
63. The Lack of Contribution of 7-Hydroxymitragynine to the Antinociceptive Effects of Mitragynine in Mice: A Pharmacokinetic and Pharmacodynamic Study
64. KCAC Position Statements - Global Kratom Coalition
65. Nonswallowed Kratom-Derived Products: Unlawful Dietary Supplements That Endanger Public Health - PubMed

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